2.30 Prove that any formula built up from $\neg$ and $\to$ in which no propositional variable occurs more than once cannot be a tautology.

Proof. We proceed through induction on the length of a formula.

Base Case: A formula of length zero is just composed of a single propositional variable, call is p. It is clear that the truth table for p has two rows where the first truth value is T while the second is F. Thus a formula of length zero is not a tautology.

Inductive Hypothesis: All formulas built up from $\neg$ and $\to$ in which no propositional variable occurs more than once of length less than or equal to n are neither tautologies nor contradictions. That is, there exists one row in the truth table which gives the described a truth value of F.

Inductive Step: Let ϕ be a formula of length $n + 1$. Thus ϕ has a unique principal connective so it of one of these forms: $\neg\theta$ or $\theta \to \psi$.

Case $\neg\theta$: Suppose $\phi = \neg\theta$. Then θ has length less than or equal to n. By the inductive hypothesis the row(s) which made θ T are flipped to F and vice-versa.

Case $\theta \to \psi$: Suppose $\phi = \theta \to \psi$. So both θ and ψ have length less than or equal to n hence inductive hypothesis applies to them. Since both θ and ψ do not share any propositional variables, there is a row in the truth table in which both θ is true and ψ is false making ϕ false. Similarly, all the other rows of the truth table would make ϕ true.

□